rf-de-20m — Design Document

ai-ee v1 pipeline

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Contents

1 Board and Run Metadata

Field	Value
board	rf-de-20m
workspace	`boards/rf-de-20m`
phase	P4
gate status	no gates recorded yet
state created	2026-08-07T20:58:16
state updated	2026-08-08T00:22:50

2 Overview

BRIEF - rf-de-20m: 20 MHz Class E GaN RF power stage, 200 W

Owner-approved 2026-08-07. Full derivation + rejected alternatives: `LEARNINGS.md` (this dir). **Read that file before changing any number here** - the operating point is frozen and the Class DE / gate-drive-transformer alternatives are already closed.

Scope (deliberately narrow - owner asked for power stage ONLY)

IN scope: gate driver, GaN power stage, output filter + impedance match, 5 V housekeeping. OUT of scope: no MCU, no PWM generation, no telemetry, no protection logic, no enclosure. PWM arrives from an external signal generator via SMA. This is a bench power stage.

Hard constraints

- **100% JLCPCB PCBA - zero off-catalog parts.** Owner ruled on this explicitly; it is the reason the topology is Class E and not the originally requested Class DE (no LCSC-stocked driver can switch a 20 MHz half-bridge; see LEARNINGS). Every part must be LCSC/JLC stocked.
- 4-layer PCB. L2 = unbroken GND directly under the power loop. Non-negotiable at 20 MHz.
- Minimise spend by reducing SCOPE, never by reducing quality.

Operating point (frozen - do not re-derive)

```
| Param | Value |
|---|---|
| Frequency | 20 MHz |
| Output power | 200 W into 50 ohm |
| Bus Vdd | 40 V DC (needs >=6 A supply) |
| R_opt (Class E) | 4.614 ohm |
| Vds peak | 142.5 V (3.562 x Vdd), 1.4x margin on 200 V part |
| Ids peak | ~16 A |
| I_dc | ~5.8 A |
| Expected efficiency | **80-85%** (NOT 86% - see thermal note) |
```

Core parts (both LCSC-verified in stock)

- **Switch:** EPC2019 eGaN FET - 200 V, 36 mohm, Coss 110 pF, Qoss 18 nC@100 V. **LCSC C2836675**, ~$2.17. 1.35 mm chip-scale LGA, bottom-cooled through its solder balls.
- **Driver:** LMG1020YFFR - 5 V, 7 A source / 5 A sink, 3 ns prop delay, 1 ns min pulse, 60 MHz. **LCSC C6423790**, ~$0.36, 0.8x1.2 mm WCSP. NB: LCSC brands it "Tokmas", not TI - flag for authenticity check on receipt.
- **5 V rail:** small 40 V-input buck. NOT an LDO - gate charge draw is Qg*f ~ 0.1 A, so an LDO would burn (40-5)*0.1 = 3.5 W. Pick an LCSC-stocked 40 V-capable buck.

Output network (Class E, Q_L = 5)

Drain –+– C_shunt –GND C_shunt 317 pF TOTAL (Coss 110 pF + ~200 pF ext C0G)

|

+– L_s – C_s –+– L_m –+– SMA out (50 ohm)

| |

(–) C_m – GND
L_s 184 nH | C_s ~447 pF | L_m ~115 nH | C_m ~500 pF (match Q 3.14, 4.614 -> 50 ohm)

MUST-RESOLVE during build (do not silently skip)

1. **Tank current is 6.6 A rms.** In a series-resonant tank the circulating current equals the load current - loaded Q sets VOLTAGE across L/C, not current. Size L and C by *dissipation*, not current rating. 2. **The output inductor is the hottest component on the board** (~7 W at Q=150), hotter than the GaN. If no LCSC RF inductor carries 6.6 A rms at high Q: parallel several (splits current AND ESR), or use a PCB air-core spiral (free, JLC-native, costs area). 3. **Parallel several C0G caps** for C_s / C_shunt to split ripple current. L/C see ~215 V peak -> specify >=250 V, ideally 500 V C0G. No X7R anywhere in the tank (voltage/temp coefficient). 4. **Verify LMG1020 input** accepts a 50 ohm-terminated logic-level SMA signal directly; add bias / a fast buffer if not. 5. Total dissipation 35-50 W. Thermal path for the EPC2019 is vias straight into plane copper.

Layout rules

- Gate loop (LMG1020 -> EPC2019 gate -> source) must be the tightest loop on the board. The WCSP driver exists to be placed millimetres from the FET; do not waste that.
- Power loop area, not trace width, sets the ringing that the 1.4x Vds derate protects.
- Skin depth in Cu at 20 MHz is 14.6 um; 1 oz = 35 um is only ˜2.4 skin depths. Pour tank and power-loop conductors wide - do NOT size by DC current.
- Decoupling: multiple small-case C0G right at the drain/source. Bulk electrolytic is useless at 20 MHz - it is there only for the 5.8 A DC feed.
- RF choke feeding the drain must be a real RF choke, self-resonance well above 20 MHz.
- Keep the 50 ohm output trace controlled-impedance to the SMA.

3 Requirements

requirements.md - rf-de-20m

20 MHz Class E GaN RF power stage, 200 W into 50 ohm. Source: `brief/BRIEF.md` + `brief/RESEARCH-LEARNINGS.md`, owner-approved 2026-08-07.

Frozen inputs - not open for re-derivation. The topology, operating point and the two core parts below were decided by the owner after a full costed trade study. They are recorded here as settled requirements, not as proposals. Class DE was evaluated and REJECTED (no LCSC-stocked gate driver can switch a 20 MHz half-bridge). Do not reopen.

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1. Function

A bench RF power amplifier stage: a single-ended Class E inverter that converts a 40 V DC bus into 200 W of 20 MHz RF power delivered to a 50 ohm load. An externally generated 20 MHz PWM drive signal arrives on an SMA input, is buffered by a low-side GaN gate driver, and switches a single ground-referenced eGaN FET. The FET's drain feeds a shunt capacitance plus a series resonant tank and an L-match that transforms the 4.614 ohm Class E load line up to 50 ohm at the output SMA. Scope is the POWER STAGE ONLY. There is no MCU, no on-board PWM generation, no telemetry, and no protection logic on this board.

2. Interfaces

```
| # | Interface | Direction | Standard / detail | Connector |
|---|---|---|---|---|
| I1 | RF drive input | in | 20 MHz logic-level PWM from an external signal
generator; 50 ohm terminated at the board.  Feeds the LMG1020 input.
Adjustable duty cycle expected (~50% nominal for Class E). | SMA (stated in
brief) |
| I2 | RF power output | out | 20 MHz, 200 W CW into 50 ohm.
Controlled-impedance 50 ohm trace from the L-match to the connector.  | SMA
(stated in brief) - see Q10 |
| I3 | DC bus input | in | 40 V DC, >=6 A. | NOT STATED - see Q5 |
| I4 | 5 V housekeeping | in or internal | 5 V rail for the LMG1020.  Source not
yet decided:  onboard 40 V-input buck vs external 5 V feed.  | see Q9 |
```

Notes carried from the brief that bind the interfaces:

- The 50 ohm output trace must be controlled-impedance all the way to the output connector.

- MUST-RESOLVE at design time: verify the LMG1020 input accepts a 50 ohm-terminated logic-level SMA signal directly; add bias or a fast buffer if it does not.
- No other external interfaces exist. No status LEDs, no buttons, no debug/telemetry header are required by the brief. ASSUMED: none wanted (scope is deliberately bare bones).

3. Power

Main bus (frozen):

Param	Value	Status
Bus Vdd	40 V DC	frozen
Bus current, DC average	~5.8 A at 85% efficiency	frozen
Supply requirement	>=6 A capable at 40 V	frozen
Output power	200 W into 50 ohm	frozen
Class E load line R_opt	4.614 ohm	frozen
Vds peak	142.5 V (3.562 x Vdd) -> 1.4x margin on the 200 V EPC2019	frozen
Ids peak	~16 A	frozen
Tank current	6.6 A rms (equals load current - loaded Q sets VOLTAGE across L/C, not current)	frozen
Expected efficiency	80-85% (NOT 86%)	frozen
Total dissipation	35-50 W, most of it in the passives, not the semiconductor	frozen

Housekeeping rail:

- 5 V for the LMG1020 gate driver. Must be a buck, NOT an LDO: gate-charge draw is Qg*f ~ 0.1 A, so a 40 V -> 5 V LDO would burn (40-5)*0.1 = 3.5 W.
- GUESS: 5 V rail budget 150-250 mA (0.1 A gate charge + driver quiescent + margin). To be confirmed against the actual EPC2019 Qg at the chosen drive voltage during design.
- If an onboard buck is used it must be LCSC-stocked and rated for a 40 V input (see Q9).

Frozen part selections:

- Switch: EPC2019 eGaN FET, 200 V / 36 mohm / Coss 110 pF / Qoss 18 nC @ 100 V. LCSC C2836675, ~$2.17. 1.35 mm chip-scale LGA, bottom-cooled through its solder balls.
- Driver: LMG1020YFFR, 5 V, 7 A source / 5 A sink, 3 ns prop delay, 1 ns min pulse, 60 MHz. LCSC C6423790, ~$0.36, 0.8 x 1.2 mm WCSP. NB: LCSC brands it "Tokmas", not TI - flag for an authenticity check on receipt.

Frozen output network (Class E, Q_L = 5):
Drain –+– C_shunt –GND C_shunt 317 pF TOTAL (Coss 110 pF + ~200 pF external C0G)

|

+– L_s – C_s –+– L_m –+– SMA out (50 ohm)

|

C_m – GND
L_s 184 nH | C_s ~447 pF | L_m ~115 nH | C_m ~500 pF (match Q 3.14, 4.614 -> 50 ohm)
Binding design constraints on that network (from the brief's MUST-RESOLVE list):

- The output inductor is the hottest component on the board (~7 W at Q=150), hotter than the GaN. Size L and C by DISSIPATION and Q, not by current rating. If no LCSC RF inductor carries 6.6 A rms at high Q: parallel several (splits current AND ESR), or use a PCB air-core spiral (JLC-native, free in BOM, costs board area).
- Parallel several C0G caps for C_s and C_shunt to split ripple current. Tank L/C see ~215 V peak -> specify >=250 V, ideally 500 V C0G. No X7R anywhere in the tank.
- The drain-feed RF choke must be a real RF choke with self-resonance well above 20 MHz.
- Decoupling: multiple small-case (0402/0603) C0G right at drain/source. Bulk electrolytic is useless at 20 MHz and exists only to support the 5.8 A DC feed.

4. Environment

NOT STATED in the brief. What is known:

- Bench instrument, no enclosure (explicitly out of scope). Ingress protection: none required.
- Vibration: none - stationary bench use. ASSUMED.
- Ambient temperature: not stated - see Q7.
- Cooling: 35-50 W of dissipation must leave the board. The EPC2019 is bottom-cooled through its solder balls, so its thermal path is vias straight into plane copper. Whether that copper is then coupled to a heatsink and whether forced air is present is NOT STATED - see Q6.
- Duty cycle (continuous 200 W vs pulsed/keyed) is NOT STATED and materially changes the thermal design - see Q8.

5. Size & mounting

NOT STATED in the brief - no outline, no mounting holes, no height limit given. See Q3 and Q4. Constraints that are already fixed and will bound whatever outline is chosen:

- 4-layer stackup is MANDATORY (non-negotiable at 20 MHz). L2 must be an unbroken GND plane directly under the power loop.
- Board area is a design variable, not just a cost item: the fallback for the output inductor is a PCB air-core spiral, which trades board area for BOM cost and sourcing risk. A tight HARD outline cap could remove that fallback.
- Copper pours for the tank and power loop must be wide (skin depth in Cu at 20 MHz is 14.6 um; 1 oz = 35 um is only ~2.4 skin depths), so conductor area cannot be shrunk to DC-current sizing.

WARNING: any outline cap answered as HARD binds permanently at P5 board_init and cannot be relaxed later without restarting layout. Answer HARD only if it is a real physical constraint.

6. Quantity & budget

NOT STATED in the brief - see Q1 and Q2.

The only budget guidance given is directional: "minimise spend by reducing SCOPE, never by reducing quality." Known BOM anchors: EPC2019 ~$2.17, LMG1020YFFR ~$0.36.

7. Assembly

- **100% JLCPCB PCBA. Zero off-catalog parts.** Owner ruled on this explicitly and it is the reason the topology is Class E rather than the originally requested Class DE. Every part must be LCSC/JLC stocked. This is a HARD constraint.
- No hand-soldered parts, no wound magnetics, no THT-only parts. (This is what rejected the gate drive transformer, the push-pull output transformer, and Mini-Circuits/Coilcraft RF transformers.)
- 4-layer PCB.
- Assembly sides: NOT STATED. ASSUMED single-sided (top) assembly, because the EPC2019 is bottom-cooled into plane copper and a clear bottom side is useful for heatsinking - confirm via Q6.
- The two connectors (I1, I2) and the DC input (I3) must be JLC-assemblable parts, which constrains the connector choices in Q5 and Q10.

8. Compliance / safety flags

All three of the following APPLY to this board.

1. **>30 V present (40 V bus, 142.5 V peak switch node).** The DC bus is 40 V and the drain node swings to a designed 142.5 V peak, with real ringing on top of that - the 1.4x derate against the EPC2019's 200 V rating is the entire margin. Consequences: creepage/clearance on the drain net and tank must be sized for >=215 V peak (tank L/C see ˜215 V pk); the 1.4x derate is protected by POWER LOOP AREA, not trace width, so layout is a safety item here, not just performance. Exposed high-voltage nodes on an unenclosed bench board are a shock/arc hazard. 2. **High current (>3 A): ˜5.8 A DC input, ˜16 A peak switch current, 6.6 A rms tank current.** Conductor sizing must account for skin effect at 20 MHz (14.6 um skin depth), not DC ampacity. The output inductor dissipates ˜7 W and is the hottest component on the board - a burn hazard on an open bench board and a fire/derating risk if the chosen part is sized by current rating instead of dissipation. 3. **RF transmit: 200 W at 20 MHz.** This is a high-power HF RF source in a band adjacent to the 13.56 MHz and 27.12 MHz ISM allocations; 20 MHz itself is NOT an ISM allocation. EMC implications: a Class E stage is a hard-switched square-wave source, so harmonics at 40, 60, 80 MHz+ are strong and the only suppression is the output tank/match. Unenclosed, unshielded operation at 200 W will radiate. This board is for operation into a matched dummy load or a properly licensed/shielded setup only - it is not a certifiable transmitter and must not be connected to an antenna without an appropriate licence and additional filtering. 4. **Load-sensitivity hazard (consequence of the frozen topology).** Class E is narrowband and load-sensitive; it does not maintain ZVS across load pull. With protection logic explicitly out of scope, an open, shorted, or badly mismatched output at 200 W can drive Vds past the 1.4x margin and destroy the FET. This needs an explicit owner acknowledgement - see Q11.

Not applicable: no mains voltage anywhere on this board (external 40 V bench supply); no battery of any chemistry; no motors.

9. Open questions

Answer these and the design can proceed. Defaults are offered where a sensible one exists - if a default is fine, just say "default".

1. **How many boards do you want built?** (Default: 5 - JLCPCB's typical minimum-cost quantity for a 4-layer assembled board.)

2. **What is your target cost per board, assembled?** A rough ceiling is enough (e.g. "under $60 each"). This mainly decides whether the output inductor is bought (several paralleled SMD parts) or etched as a free PCB spiral that costs board area. (Default: no explicit ceiling - optimise for quality within a sensible bench-instrument cost.)

3. **Is there a maximum board size?** If yes, give width x height in mm, and say whether that limit is **HARD** (a real physical constraint, e.g. it must fit an existing fixture - this binds permanently and cannot be changed later) or **soft** (a preference). (Default: soft guidance of about 100 x 80 mm, no hard cap.)

4. **Do you want mounting holes?** If yes: how many, what screw size, and are their positions fixed by something existing? (Default: 4x M3 holes, one near each corner, positions chosen by the layout.)

5. **How should the 40 V DC bus connect to the board?** It must carry ~6 A continuously. Options: (a) a screw terminal block, (b) a high-current 2-pin header, (c) solder lugs/pads for ring-terminal wires, (d) a banana-jack pair. Also: do you already have a 40 V supply that can deliver 6 A, and what does its output cable terminate in? (Default: (a) a 2-position screw terminal rated >=10 A.)

6. **How will the board be cooled?** It dissipates 35-50 W. Options: (a) bare board on the bench with no heatsink (this will get very hot and may not be survivable at continuous 200 W), (b) a heatsink bolted to the bottom of the board under the GaN FET, (c) a fan blowing across the top of the board, (d) both (b) and (c). Your answer decides whether the bottom copper must be left clear and flat, so it also fixes whether assembly is single- or double-sided. (Default: (d) - design for a bottom-side heatsink plus forced air, since that is the only answer that makes continuous 200 W comfortable.)

7. **What ambient (room) temperature should the board be designed to survive?** (Default: 25 C typical bench, designed with margin to 40 C.)

8. **Will this run at 200 W continuously, or in short bursts?** If bursts: roughly how long is a burst and how often (e.g. "10 seconds on, a minute off")? This changes the thermal design substantially - a pulsed duty cycle can make the hot output inductor a non-issue. (Default: continuous 200 W - the worst case, and the safe assumption.)

9. **Where should the 5 V driver supply come from?** Options: (a) an onboard step-down converter running off the 40 V bus, so the board needs only one power connection; (b) a separate 5 V input you feed from a bench supply or USB brick, which is one more cable but removes a part and removes 40 V switching noise near the gate drive. (Default: (a) onboard buck - one cable is worth more on a bench instrument.)

10. **Confirm SMA for the 200 W RF output.** The brief says SMA and SMA is electrically fine at 20 MHz, but SMA is a small connector for 200 W and repeated mating at that power is a known wear point; a BNC or N-type is more robust. (Default: keep SMA as briefed.)

11. **SAFETY - please confirm explicitly:** this board has NO protection of any kind (no VSWR detection, no overcurrent, no overvoltage, no thermal shutdown - all were placed out of scope). Class E is load-sensitive, so running it into an open, a short, or a badly mismatched load at 200 W can push the switch node past its voltage margin and destroy the FET, and the RF output is a burn/RF-exposure hazard at 200 W. Do you confirm this will only ever be operated into a proper 50 ohm dummy load or matched load, on a bench, by you? (No default - this must be answered.)

12. **What does your signal generator actually put out on the drive SMA?** Specifically: peak-to- peak amplitude, whether it can drive a 50 ohm load, and whether its duty cycle is adjustable at 20 MHz. This decides whether the LMG1020 can be driven directly or needs a bias network or a fast buffer in front of it. (Default: assume a 50 ohm-capable generator putting out ~5 Vpp into 50 ohm with adjustable duty cycle, and design the input to tolerate 3.3-5 Vpp.)

—

10. ANSWERS - all 12 questions closed by the owner 2026-08-07 (P0 batch)

These are settled. Downstream phases treat them as requirements, not proposals.

```
| # | Answer |
|---|---|
| 1 | Build quantity **5** (default taken) |
```

```
| 2 | **No hard cost ceiling**; minimise spend by reducing scope, never quality (default taken) |
| 3 | Outline **~100 x 80 mm, SOFT - no hard cap** (default taken). Deliberately NOT hard-capped so the PCB air-core spiral inductor fallback stays reachable |
| 4 | **4x M3** mounting holes at corners (default taken) |
| 5 | 40 V bus via **2-position screw terminal, >=10 A** (default taken) |
| 6 | **OWNER: continuous 200 W + bottom-side heatsink + forced air.** Bottom copper stays clear and flat as a heatsink face -> **single-sided (top) assembly** |
| 7 | Ambient **25 C bench, margin to 40 C** (default taken) |
| 8 | **OWNER: continuous 200 W.** Worst case; thermal drives the layout |
| 9 | **5 V from an onboard buck** off the 40 V bus - one cable (default taken) |
| 10 | **OWNER: keep SMA.** 100 Vrms / 2 Arms at 20 MHz is within SMA HF limits; N-type would break the 100% JLC PCBA rule |
| 11 | **OWNER SAFETY ACK: confirmed.** No VSWR / OCP / OVP / thermal protection of any kind.  Operated into a proper 50 ohm matched or dummy load ONLY. Owner accepts that an open, short or bad mismatch at 200 W will likely destroy the FET |
| 12 | **OWNER: ~5 Vpp into 50 ohm, duty adjustable.** -> 50 ohm termination direct into the LMG1020 input, **NO buffer stage**.  Closes brief MUST-RESOLVE #4 |
```

Consequences that bind later phases

- **Single-sided top assembly.** Bottom layer is reserved as a flat heatsink mounting face; no bottom-side components. Thermal via array under the EPC2019 down to bottom copper.
- **Thermal is a first-class layout driver**, not an afterthought: 35-50 W continuous, and the output tank inductor (~7 W) runs hotter than the GaN.
- **No protection parts** are to be added by any later phase. If a reviewer flags the missing protection, the response is "waived, owner-acknowledged at P0" - not a new part.
- **Outline stays soft** through P5 `board_init`; do not pass a hard `--outline` cap that would foreclose a PCB spiral inductor.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Thermal: continuous 200 W duty, bottom-side heatsink pad + thermal via array under EPC2019, forced air assumed; single-sided (top) assembly so bottom is a heatsink mounting face	Owner answer at P0 batch. 35-50 W loss with output inductor ~7 W hotter than the GaN; worst-case design chosen
P0	SAFETY ACK: no VSWR/OCP/OVP/thermal protection; matched or dummy 50 ohm load ONLY. Open/short output at 200 W will likely destroy the FET	Owner explicitly acknowledged. Class E loses ZVS under load pull; drain exceeds the 142.5 V design peak. Protection stays out of scope per power-stage-only brief

Phase	What	Why
P0	RF output connector = SMA (edge/PCB mount, 50 ohm, LCSC-stocked)	200 W into 50 ohm at 20 MHz = 100 Vrms / 2 Arms, within SMA HF limits. N-type would break the 100% JLC PCBA rule
P0	Drive input: 50 ohm termination direct to LMG1020 input, NO buffer stage	Owner gen supplies ˜5 Vpp into 50 ohm with adjustable duty. Closes brief MUST-RESOLVE #4; fewest parts and lowest added delay
P0	Defaults taken (owner did not specify): qty 5; no hard cost ceiling; outline ˜100x80 mm SOFT (no hard cap); 4x M3 corner holes; 40 V bus via 2-pos screw terminal >=10 A; ambient 25 C bench w/ margin to 40 C; 5 V rail from onboard buck off the 40 V bus	Routine judgement calls. Outline deliberately left SOFT so the PCB air-core spiral inductor fallback stays available for the 6.6 A rms tank
P1	CORRECTED EPC2019 specs from datasheet: Coss(er) 156 pF (not 110), package 2.77x0.95mm solder bars (not 1.35mm LGA), Rds(on) 22 typ/42 max (not 36), Qg 2.4/2.9 nC	Brief figures came from a web summary, not the datasheet. Two P1 agents independently verified. External C_shunt drops to ˜161 pF
P1	L_s and L_m to be etched as PCB air-core spirals, NOT catalog inductors	No LCSC part closes it: power inductors give Q 20-40 at 20 MHz (25-50 W); RF inductors rated 120-140 mA vs 6.6 A needed. Paralleling does NOT reduce loss (Q_total = Q_each). Spiral gives Q˜130-150 at zero BOM cost
P1	Two-zone floorplan mandatory: heatsink zone (Q1/L1/bulk/driver) vs magnetics zone (L_s/L_m/tank caps) with L2 plane keepout under spirals only	A metal heatsink or solid ground plane under a PCB spiral is a shorted turn. L2 stays unbroken under the power loop; cut out ONLY under spirals
P2	TWO EPC2019 in parallel (overrides the frozen single-FET brief)	H1 owner approval. Only config where thermal closes: Tj 103 C vs 138 C. 2x Coss(er)=312 pF meets the 317 pF shunt need, deleting the external cap. 2-FET layout can build with one populated; reverse is impossible
P2	Drive input is unipolar 0/+5V (generator TTL/CMOS mode); DC-coupled, no level shift, no AC coupling	H1 owner answer. LMG1020 input abs max -0.3 V forbids bipolar swing; AC coupling would destroy duty-cycle control that Class E depends on
P2	Controlled-impedance 50 ohm output trace RELAXED to a wide low-loss pour	H1 owner approval. At 20 MHz a 15 mm run is lambda/550, perturbs load <2%; the controlled-Z 0.4332 mm geometry would run 50-75 C at 2 A rms
P2	Outline 100x80 mm, P6 PRE-AUTHORISED to grow to 120x80 if a spiral cannot reach area. No hard –outline cap at P5	H1 owner approval. Spiral Q drives the dominant 16 W magnetics loss; ˜+$3-6/board is worth protecting it
P3	EPC2019 (C2836675) is OUT OF STOCK at LCSC (stock=0, price up 2.17->3.93). Design CONTINUES; order held pending restock. Re-verify at P10	Verified twice independently. No in-stock GaN substitute exists: 650V parts have Coss 239-760pF and 110-600mohm (8-11W conduction); EPC2051/2036 are 100-120V forcing a 25-33V bus. Stock-out does not change footprint/pinout/schematic, and it was in stock this morning, so design work is not wasted

Phase	What	Why
P3	LMG1020YFFR primary = Tokmas C6423790; alternate = genuine TI-branded C506817 (in stock)	Part-sourcer found a real TI listing, resolving the architecture's authenticity flag
P3	RF choke L201 = 2x FXL0630-R47-M (470nH, 4.1mohm, 20A) in SERIES	No single LCSC part meets L/SRF/DCR/Isat together (SRF is essentially never published for adequately-rated parts). Uses the architecture's pre-authorised escape
P2	AMENDED operating point (Sokal finite Q_L=5 coefficients): R_opt 4.13 ohm, C_shunt 403 pF, L_s 164 nH, C_s 518 pF, L_m 110 nH, C_m 530 pF, I_tank 6.96 A rms. Bus STAYS 40 V	Frozen brief used 1/(wR*5.4466), which is Sokal's Q_L->infinity constant, but the board runs Q_L=5. Loss 38.6 W / 83.8%, inside the frozen band
P2	External shunt trim cap POPULATED at 3x33 pF (~87 pF), not DNP	EPC2019 Coss spreads 110-150 pF (+36%). If device Coss is the whole shunt, that spread lands on ZVS with no absorber. The external cap is the only trim element
P2	TWO EPC2019 now MANDATORY (not merely preferred): single FET reaches Tj 160 C with a hypothetical 0 C/W heatsink, above the 150 C abs max	Real Rds(on) 36/50 mohm (65/90 hot). Pair gives 114 C nominal / 133 C max-corner at theta_HS <= 0.7 C/W. Dissipation is flat N=1->2; the win is RthJB halving per die
P3	SMA re-sourced to C22418168 CONSMA001-SMD-G-T (TE/Linx), verified TRUE SMD from manufacturer drawing; 100% SMT preserved. Cost 0.39 -> 2.97 each	Previous BWSMA-KE-P001 was a THT vertical panel jack despite an SMD label. LCSC package/type fields are unreliable for mount style; verified via pad-layer inspection + customer drawing
P3	EPC2019 footprint: end columns are 0.45 mm c-c (corrected 0.46 -> 0.450). The 0.68-0.70 mm figure is the OUTER ENVELOPE and was NOT applied	Fixer extracted datasheet vector geometry and refuted the librarian report. Applying 0.68 would have shifted the GATE pad 0.11 mm off a 0.25 mm pad

Sheet plan

rf-de-20m - hierarchical sheet plan

AMENDED 2026-08-07 (P2-A): part values and net currents re-derived at the corrected operating point (**R = 4.13 ohm**, from Sokal's Q_L = 5 coefficients) on the real EPC2019 datasheet. The sheet split, refdes blocks and net-naming contract are unchanged. See `blocks.md` s0 and `decisions.md` D11.

Three sheets under a root that contains nothing but the sheet symbols and the inter-sheet wiring. ~60 parts. **Deliberately not decomposed further** - this is one function (a Class E stage) and the only organising principle that earns its keep is the floorplan, so the sheets map 1:1 onto the zones in `stackup.md` s3.

```
| Sheet | Blocks (‘blocks.md‘) | Zone | Refdes block | ‘#PWR‘ base |
|---|---|---|---|---|
| **‘hk‘** | B1 bus entry + bulk, B2 +5 V buck | A | **100-199** | **100** |
| **‘stage‘** | B3 drive input, B4 gate driver, B5 FET pair, B6 drain feed + bus
decoupling | A | **200-299** | **200** |
| **‘tank‘** | B7 series tank, B8 L-match, B9 RF output | B and C | **300-399** |
**300** |
```

Refdes are unique across sheets by construction: each sheet owns a hundreds block for every prefix, and each sheet has its own #PWR base.

—

1. Net-naming contract - BINDING ON P4

KiCad names a net from a hierarchical label alone as **‘/<sheet>/<LABEL>‘**. A net that lives entirely inside one sheet therefore comes out with the sheet prefix, and `constraints.json` uses those names. **A net whose name does not match is silently no-op'd by every P5-P8 consumer** - this exact trap cost the lumina-par run a P4 amendment.

Power nets are **bare global symbols**: `+40V`, `+5V`, `GND`.

Exactly one signal net crosses a sheet boundary: ‘/SW‘. For it to be named `/SW` and not `/stage/SW`, **P4 must place a root-sheet local label spelled `SW` on the inter-sheet wire.** Everything else is sheet-internal and is named with its sheet prefix on purpose.

Canonical net	Scope	Carries
‘+40V‘	global	**5.96 A** DC nominal, 7.0 A declared. Ring to 51 V
‘+5V‘	global	**99 mA** avg (real Qg 1.8 nC), ~1.6 A/gate peaks
‘GND‘	global	In1 + In2 + B.Cu plane stack, zones A and C
‘/SW‘	**‘stage‘ -> ‘tank‘, root label required**	drain node. **9.17 A rms, 17.1 A pk, 142.5 V pk**
‘/stage/DRIVE‘	‘stage‘	J201 -> R201/R202 -> U201 IN+
‘/stage/GATE_ON‘	‘stage‘	U201 OUTH -> the two turn-on legs
‘/stage/GATE_OFF‘	‘stage‘	U201 OUTL -> the two turn-off legs
‘/stage/GATE_Q1‘	‘stage‘	gate of Q201
‘/stage/GATE_Q2‘	‘stage‘	gate of Q202
‘/tank/TANK_A‘	‘tank‘	L301 / C_s junction. **6.96 A rms, 156 V pk - the highest node on the board**
‘/tank/TANK_B‘	‘tank‘	C_s / L302 junction. 6.96 A rms, 41 V pk
‘/tank/RFOUT‘	‘tank‘	L302 -> C_m -> J301. 6.96 A rms at the C_m node, 2.0 A into the load, 141 V pk

The OUTH/OUTL nets are named ‘_ON‘/‘_OFF‘, NOT ‘_H‘/‘_L‘, on purpose. `rules_gen.detect_diff_pairs` auto-pairs `high_speed` nets whose names end in _H/_L (as well as _P/_N, _DP/_DM, +/-) and would silently treat the turn-on and turn-off legs as a 100 ohm differential pair, emitting a `diff_pair_gap` rule and an inner-layer `disallow track` rule on the two most inductance-critical nets on the board. Do not rename them back.

Do not "tidy" ‘/stage/ or /tank/‘ into bare names. Promoting a sheet-internal net to a root-crossing name means the root sees wire + label + one sheet pin, which raises ERC `label_dangling`; buying the bare name would mean adding a part to the root sheet. The netlist is the authority.

—

2. `hk` (100-199, `#PWR` 100) - zone A

Ref	Part / class	Note
J101	2-pos screw terminal, 5.08 mm, >= 24 A / 250 V (KF128-5.08-2P class)	**The only THT part on the board.** Left edge, x < 5 mm, clear of the heatsink land (HS-3)
C101, C102	2x 100 uF / **63 V** SMD polymer can	bulk. SMD can, not leaded - no bottom-face solder
C103, C104	2x 2.2 uF / 100 V **X7R** 0805	mid tier, 100 kHz-5 MHz. **X7R is correct here and nowhere else on this board**
U101	**LM5017-class 100 V synchronous COT buck**, SOIC-8-EP	Vin abs max >= 63 V; no comp network, no catch diode
L101	15-47 uH shielded, >= 0.5 A	placeholder value; **P4 sizes against the chosen Fsw**
C105	Vin local, 100 V	

C106, C107	VCC decoupler, bootstrap	per the buck datasheet
C108, C109	Vout bulk + HF on `+5V`	
R101, R102	feedback divider	sets 5.0 V +/-4%
R103	RON / on-time resistor	sets Fsw

Placement rule: at the DC-input end, as far from the gate loop and the tank as zone A allows. Its 0.5-2 MHz switching noise is spectrally clear of 20 MHz but its switch node is still a dV/dt source next to a 1.4 V gate threshold.

—

3. `stage` (200-299, `#PWR` 200) - zone A

This sheet is the board. Everything on it lives in one ~15 x 15 mm cluster except the drive connector.

Ref	Part / class	Note
J201	SMD edge-launch SMA, 50 ohm (BWSMA-KE-P001 class)	**SMD, not the THT bulkhead jack** - a THT jack solders through the bottom face and breaks the heatsink land
R201, R202	**2x 100 R 0805 in parallel** = 50 ohm	~0.125 W in the termination; a single 0603 is over its rating. **DC-coupled - see `blocks.md` B3**
U201	**LMG1020YFFR**, 6-ball WCSP 0.8 x 1.2 mm	**Sits ON the mirror axis, equidistant from both gate bars.** LCSC brands it "Tokmas" - authenticity check on receipt
C201	100 nF 0402 X7R	VDD bypass, **< 0.5 mm from the VDD ball, via-in-pad return**
C202	10 nF 0201 C0G	second VDD bypass, straddling the ball
R203, R204	2x 4.0 R 0603 in parallel	OUTH -> `/stage/GATE_Q1`
R205, R206	2x 4.0 R 0603 in parallel	OUTH -> `/stage/GATE_Q2`
R207, R208	2x 4.0 R 0603 in parallel	OUTL -> `/stage/GATE_Q1`
R209, R210	2x 4.0 R 0603 in parallel	OUTL -> `/stage/GATE_Q2`
Q201, Q202	**2x EPC2019** 200 V eGaN	**MIRRORED PAIR about the U201 axis.** Both populated by default
C203-C206	4x 33 pF / 1 kV C0G 1206, **0-133 pF in 33 pF steps**	**C_shunt trim - POPULATE 3 (99 pF) at P2-A**, no longer DNP. The FET pair supplies 316 pF of the 403 pF shunt; this bank supplies the balance AND is the only absorber of the part's 110-150 pF Coss spread. **In the power loop, not on a stub**
L201	RF choke, **>= 0.82 uH, SRF >= 80 MHz, DCR <= 25 mohm, I_sat >= 12 A, >= 8 A rms**	Hardest part on the BOM. **Pre-authorised escape: 2x 0.47 uH in series.** Not authorised: a smaller choke - see `blocks.md` B6
C207-C210	4x 10 nF / 100 V C0G 0603	bus HF bank, **within 3 mm of L201's bus-side pad**
C211, C212	2x 1 nF / 100 V C0G 0603	terminates the bank's self-resonance

Four resistors per FET pair per polarity is not over-specification. Individual gate resistors are what damp the differential mode between the two gate loops; a shared resistor leaves the two gate nodes coupled through the driver output and free to oscillate. Two 0603s in parallel per leg rather than one 0805 keeps each part at **0.029 W** (the real Qg of 1.8 nC drops the pair's gate-loop energy to 0.36 W) and roughly halves the leg's parasitic inductance. See `blocks.md` s3.

The gate-loop budget is the acceptance criterion for this whole cluster, and P2-A TIGHTENED it: <= 0.48 nH per FET, matched to +/-0.1 nH. The datasheet gives Ciss 200 pF and Crss 0.7 pF, so C_GS ~ 199 pF, and EPC WP008 Eq.1 at R_G + R_src = 3.1 ohm gives 0.48 nH (the retracted Qg of 2.4 nC had implied ~350 pF and 0.84 nH). **This is the tightest layout spec on the board** - two parallel 0603s are ~0.15-0.2 nH, leaving ~0.3 nH for vias and interconnect. **Stated fallback: R_G = 3 ohm relaxes the budget to 0.84 nH** at ~+0.3 W turn-off loss and ~+1.6 C of Tj; take it consciously.

Match the loops geometrically, not the trace lengths electrically - FR4 is ~6.7 ps/mm, so length matching is not the mechanism. The +/-0.1 nH exists to damp the differential mode and equalise static sharing; **skew itself is benign** in a soft-switched topology (at turn-on the drain is at ~0 V, so an early device has nothing to hog).

Common source star. Both source bars return to **one** via cluster on the mirror axis, and the LMG1020's GND connects to the source plane **only at that star point** (TI LMG1020DS s10.1.1 - single-point ground). Separate source returns would make common-source inductance differential and imbalance the pair.

—

4. `tank` (300-399, `#PWR` 300) - zones B and C

```
| Ref | Part / class | Zone | Note |
|---|---|---|---|
| **L301** | **L_s 164 nH - etched PCB air-core spiral** | B | 2 turns, OD 30-34
mm, **~2.5-3 mm trace** (wider than before - the lower L target at the same OD
needs it, and it buys Q), **>= 1430 mm^2 at Q 100**.  **No LCSC code.  ‘PCB
feature - do not place‘.** SPIRAL-1..6 |
| C301-C309 | **9x 56 pF / 1 kV C0G 1206 = 504 pF** (target 518 pF **+/-5%**) |
B | C_s.  0.77 A rms and ~105 mW each.  **No X7R.** The +/-5% is OPEN-12:  the
P1 fragment’s C_series coefficient is refuted and SIM-4 is the arbiter - keep
it a parallel bank so it trims by depopulation |
| **L302** | **L_m 110 nH - etched PCB air-core spiral** | B | **>= 950 mm^2 at Q
100 - nearly as large as L301 despite being 67% of the inductance**, because
the constraint is dissipation area, not L (SPIRAL-3) |
| C310-C319 | **10x 1206 C0G 1 kV = 530 pF +/-3%** | C | C_m.  Carries **6.66 A
rms**, not the 2 A load current - 0.67 A each |
| **J301** | SMD edge-launch SMA, 50 ohm - **same part as J201** | C | right
edge.  100 Vrms / 2.0 A rms |
```

Why C_m sits in zone C and C_s does not. C_s is a *series* element and floats - it needs no ground reference and belongs beside the coils. C_m returns to GND and carries 6.66 A rms, so it must sit where the ground plane exists.

L301 and L302 are schematic symbols with custom footprints, not annotations. The footprint carries the spiral copper on F.Cu (and B.Cu under SPIRAL-2), a courtyard equal to the thermal footprint, and its two terminal pads. Copper that no tool knows about gets routed over, poured under and placed on top of.

—

5. Placement plan for P6

5.1 Placement groups (declared in `constraints.json.placement.groups`)

```
| Group | Anchor | Members | Purpose |
|---|---|---|---|
| ‘switch‘ | **Q201** | Q202, U201, R203-R210, C201-C212, L201 | **The whole
thing.** Power loop + gate loop + choke + bus HF in one ~15 x 15 mm cluster |
| ‘drive_in‘ | J201 | R201, R202 | 50 ohm termination at the connector, not at
the driver |
| ‘tank_ls‘ | L301 | C301-C309 | C_s bank adjacent to the L_s spiral |
| ‘tank_lm‘ | L302 | C310-C319, J301 | L_m -> C_m -> SMA, in that order, shortest
possible |
| ‘hk‘ | U101 | L101, C105-C109, R101-R103 | buck and its loop |
| ‘bus_in‘ | J101 | C101-C104 | bulk at the terminal |
```

5.2 Edges

Ref	Edge	pos	Why
J101	**left**	0.5	DC entry. **x < 5 mm so its THT pins clear the heatsink land** (HS-3)
J201	**top**	0.2	drive input, in zone A near the driver
J301	**right**	0.5	RF output, in zone C. Must be adjacent to the C_m node

5.3 Parts that must be placed by explicit `place_edit` and LOCKED before the annealer runs The annealer optimises a cost function; none of the constraints below are expressible in it.

Ref	Why locking is mandatory
Q201, Q202	mirror symmetry about the U201 axis is the whole of `blocks.md` s4.1(c). An annealer will not produce a mirror pair
U201	must sit **on** the axis, equidistant from both gate bars
R203-R210	four matched legs; two arms must be geometric mirrors
L301, L302	zone B, centre-to-centre >= 38 mm, >= 14 mm clear of the heatsink land (SPIRAL-5, SPIRAL-6)
J301	zone C, adjacent to the C_m node, so `/tank/RFOUT` stays <= 15 mm
J101	x < 5 mm (HS-3)

After L301 and L302 are placed and locked, hand-add KiCad rule areas (keepouts) over both courtyards on all four layers. No pipeline check enforces "no other copper here" and `planes_gen` has no void support. Verify geometrically at P8.

5.4 Zone assignment for P6 (board-local, translate by `outline_bbox` first)

Zone	x	Sheets	Contents
A	0 - 48	`hk`, `stage`	everything except the tank
B	48 - 88	`tank`	L301, L302, C301-C309. **Nothing else may be placed here**
C	88 - 100	`tank`	C310-C319, J301

Heatsink land on B.Cu: [**5, 10, 36, 70**], declared as a back-side keepout.

—

6. Sheet-level P4 notes

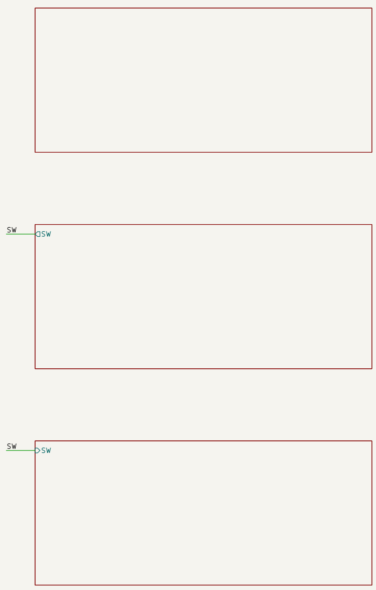
1. **The C_shunt trim bank (C203-C206) must be in the netlist, and at P2-A it is POPULATED (3 of 4 sites, 99 pF), not DNP.** The FET pair supplies 316 pF of the 403 pF shunt; the bank supplies the balance and is the **only** mechanism that absorbs the part's 110-150 pF Coss spread. The fourth site stays unpopulated as headroom. All four have pads either way, so their clearance and area are accounted for at P6/P7 and the trim stays a populate change rather than a respin. 2. **`/SW` needs a root-sheet local label** or it becomes `/stage/SW` and six `constraints.json` entries silently stop matching. 3. **Both spirals need schematic symbols.** Do not let them become a layout-only feature; their nets must be real. 4. **The drive input is DC-coupled.** Do not add a series blocking capacitor "for safety" - it destroys duty-cycle control (`blocks.md` B3). Record it on the schematic. 5. **No protection parts.** No TVS, no fuse, no clamp, no OCP, no OVP, no thermal shutdown. Owner-acknowledged at P0 Q11. If ERC or a reviewer wants one, the response is *"waived, owner-acknowledged at P0"*. 6. **L201's value is a floor, not a target.** >= 0.82 uH. If P3 cannot source it, the answer is 2x 0.47 uH in series - **not** a smaller single part. 7. **`IN-` on U201 ties to GND.** No bias divider, no buffer. 8. **Silkscreen hazard markings**: zone B (two ~130 C copper structures, one on each face if SPIRAL-2 is adopted), `/tank/TANK_A` (**156 V pk**), `/SW` (142.5 V pk), and J301 (200 W RF - burn and RF-exposure hazard). This is an unenclosed bench board with no interlocks of any kind. 9. **Silkscreen the bus voltage as a**

RANGE, not a value: "40 V max, 34-40 V". P2-A makes the bus a deliberate bring-up and derating knob (`decisions.md` D1), and ZVS is Vdd-independent, so a 36 V bench setting is a valid operating point at ~160 W - not a fault.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc`: *not recorded.* The full schematic PDF follows.



ROOT SHEET – NO PARTS BY DESIGN (sheets.md s1).

The ONLY inter–sheet signal net is /SW (the drain node, 142.5 V pk, 9.17 A rms). Its bare /SW spelling comes from the root–local labels written by add_sheet, and six constraints.json entries depend on it. Both `stage` and `tank` expose it, which is what keeps the root label connected – a root label with only ONE sheet pin raises label_dangling in KiCad 10.

GND / +40V / +5V are global POWER SYMBOLS inside the sheets and are deliberately absent here. All three PWR_FLAGs live on `hk`; a second flag anywhere else collides power_out <–> power_out.

Every other net is sheet–internal on purpose and carries its /<sheet>/ prefix. Do not 'tidy' them into bare names.

ai–ee

Sheet: /

File: rf–de–20m.kicad_sch

Title: rf–de–20m: root – 20 MHz Class E GaN stage, 200 W into 50

Size: A3	Date: 2026–08–07	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 1/4

THIS SHEET OWNS THE PWR_FLAGS for GND, +40V and +5V. Every rail enters through passive pins (J101) or a passive inductor (L101). Do NOT add a second flag on another sheet: PWR_FLAG is a power_out pin and two of them collide as power_out <-> power_out.

TYPE 3 RIPPLE INJECTION IS MANDATORY, NOT OPTIONAL. LM5017 is a COT regulator and needs >=25 mV of IN-PHASE ripple at FB (s7.3.1/s7.3.11). A 22 uF ceramic COUT gives almost none, and the regulator multi-pulses. R104/C110/C111 synthesise the ramp from the SW node so the +5 V rail keeps low ripple – which matters, because +5V feeds a driver whose VDD abs max is 5.75 V.
FB ripple 78.6 mV pk-pk at 40 V bus (3.1x the floor); Rr.Cr is 3.1x inside Eq 7.

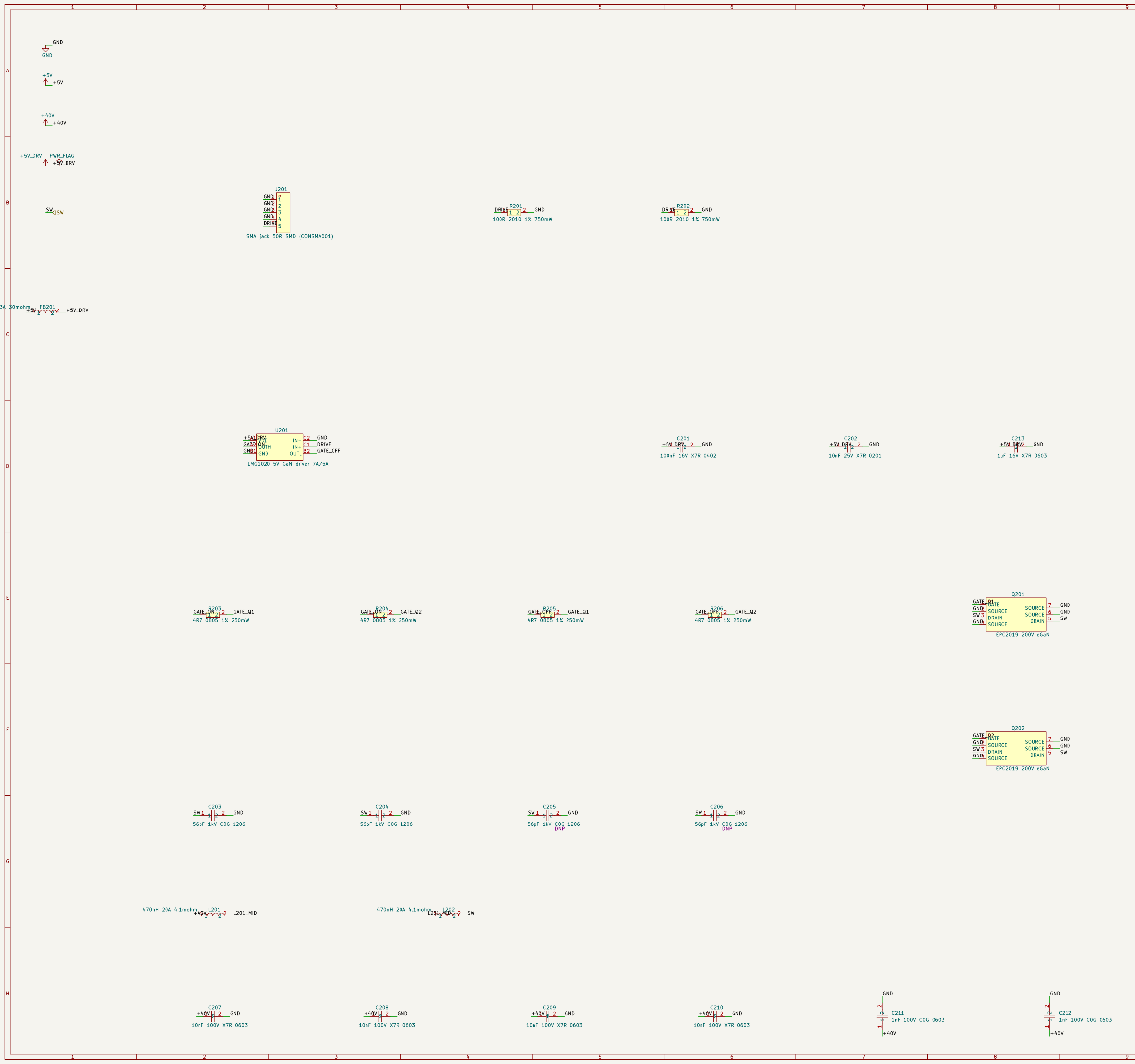
THE BUCK SWITCH NODE IS /hk/BUCK_SW. It is NOT called SW: /SW is the RF drain node and is named in six constraints.json entries.

UVLO (pin 3) IS TIED TO VIN. Datasheet s7.3.9 sanctions it explicitly; no bus-undervoltage threshold is specified anywhere in the architecture, so a divider would be an invented requirement. Abs max 100 V.

C101/C102 ARE POLARIZED: pin 1 = ANODE (+).
SETTLED 2026-08-08 against the manufacturer
drawing (JIERR RVT series, parts/C51953411
.pdf p.2): the 'Positive' leader points at
the CHAMFERED-corner end of the base, and
the footprint's two chamfers are on the
pad-1 side. The symbol's '+' at pin 1 is
therefore CORRECT. (The usual 'cut corner =
NO PROTECTION PARTS ON THIS BOARD.
cathode' rule does NOT hold on this family;
No TVS, fuse, NTC, OVP, OCP or thermal
the can's black top stripe sits on the
shutdown. Owner-acknowledged at P0 Q11.
SQUARE-cornered end. Cross-checked against
The -51 V turn-on ring is handled by
Lumimax SMDGP/RVT, same conclusion.)
VOLTAGE RATING (every +40V part >=63 V)
The footprint still carries NO silk polarity
and by MORE bulk – shrinking the bulk makes
marker – P6 MUST add a '+' outside the
the overshoot WORSE (blocks.md B1).
10.46 mm body square (the base hides it) and
an F.Fab body outline with a marker.

ai-ee
Sheet: /hk/
File: hk.kicad_sch
Title: rf-de-20m: hk – 40 V bus entry, bulk, +5 V buck

Size: A2	Date: 2026-08-07	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 2/4



EPC2019 PIN 7 IS SUBSTRATE AND IS TIED TO SOURCE. The datasheet states it three times. It is NOT expressible in the footprint – this schematic is the only place it is enforced. All SOURCE pins (2/4/6/7) -> GND; both DRAIN pins (3/5) -> /SW.

OUTH AND OUTL ARE SEPARATE OUTPUTS. OUTH sources only and floats when idle; OUTL sinks only and is held low otherwise. NO steering diode is needed or wanted. TI'S >=2 OHM MANDATE IS AT THE PIN, AND TWO FETs SHARE EACH PIN, so each branch must be >= 2 x 2 = 4 ohm. ONE 4R7 0805 per leg per FET -> 4.7/2 = 2.35R at OUTH and at OUTL (2.33R at −1%), peak drive 2.13 A. Do NOT parallel these back down.

IN− IS TIED TO GND, DRIVE GOES TO IN+. Truth table: only (IN−=L, IN+=H) drives the gate high. IN− has an internal pull-UP to VDD, so floating it holds the stage OFF; IN+ has an internal pull-DOWN, so an open drive input fails SAFE. TI's ground-bounce alternative (drive IN−, IN+ to VDD) is NOT used: it inverts, and duty cycle is the primary ZVS trim.

THE DRIVE INPUT IS DC-COUPLED – RULING D8. Do not add a series blocking capacitor 'for safety': AC coupling's DC restore pins the average at the bias point and at D=0.4 both logic levels sit above the 1.4 V threshold, so the FETs never turn off. Generator MUST be unipolar 0 to +5 V – the input abs max is −0.3 V to VDD+0.3 V, so +/−2.5 V bipolar violates it (OPEN-1).

C203–C206 ARE 56 pF, NOT 33 pF. No 33 pF part exists on this BOM; parts.json maps these sites onto the same 56 pF 1 kV C0G part as the tank banks. Populate count re-solved: 2 x 56 = 112 pF external, giving 316 + 112 = 428 pF against a 403–449 pF requirement – mid-band. C205/C206 are DNP headroom; the bank EMPTIES for a max-Coss pair, which is the point of it existing.

GATE LOOPS: aim for <= 0.48 nH per FET, matched +/−0.1 nH, geometrically MIRRORED about the U201 axis. The CRITICAL-DAMPING budget is now 1.70 nH (EPC WP008 Eq.1 at R = 4.7 + 0.4 + 0.75 = 5.85 ohm, C_GS 199 pF) – 3.6x looser than the 0.48 nH solved at the old 3.1 ohm, which is what pays for the 0805 bodies. Keep aiming at 0.48 nH. Do NOT length-match electrically – FR4 is 6.7 ps/mm and skew is benign in a soft-switched topology; matching damps the differential mode and equalises sharing.

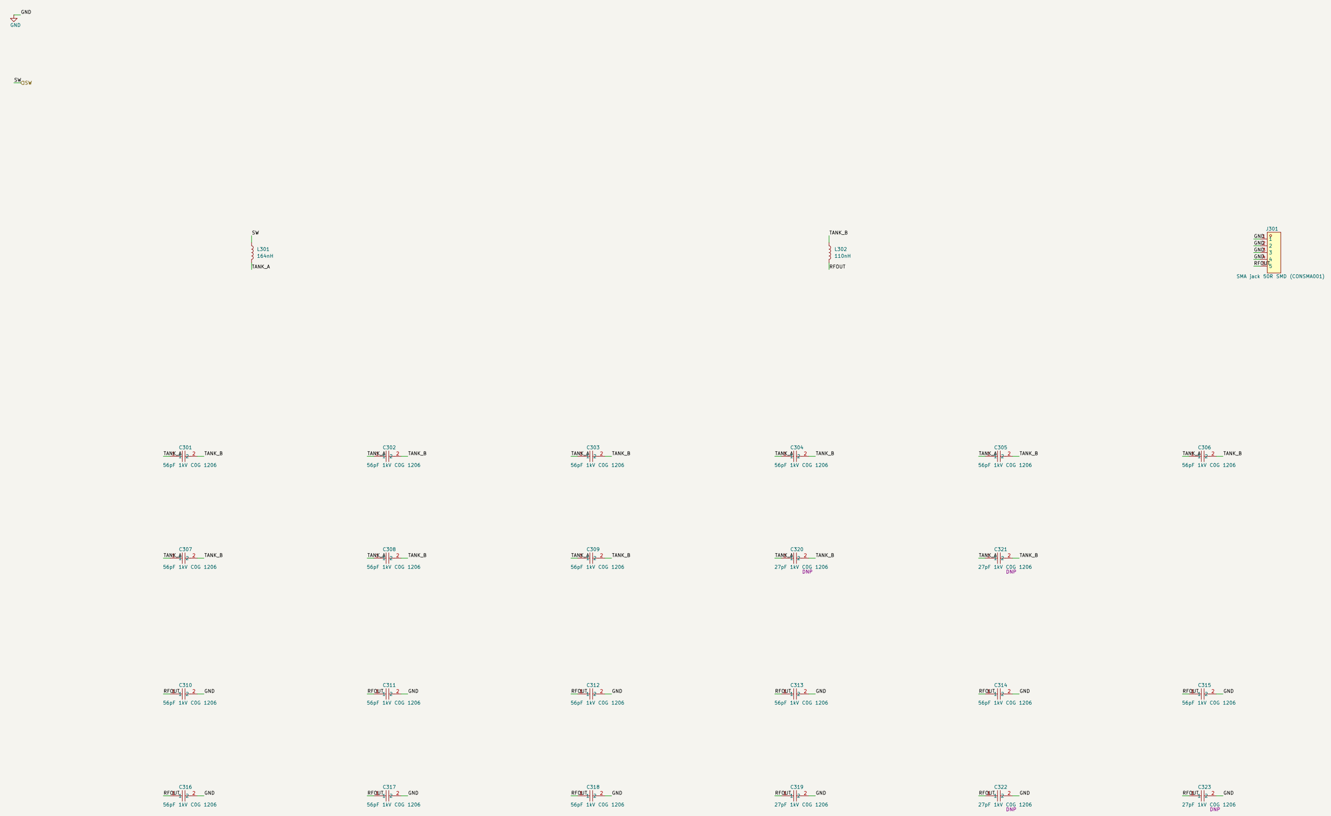
L201 + L202 ARE ONE CHOKE, IN SERIES. 0.94 uH total – the architecture's own pre-authorised escape. A smaller single choke is NOT authorised: it breaks the ideal Class E equations and RAISES the required shunt capacitance.

NO PROTECTION PARTS. No TVS on /SW, no gate clamp, no negative bias. Owner-acknowledged at P0 Q11. The 1.40x derate on a 200 V part is protected by POWER LOOP AREA at P6/P7. FB201 IS A FILTER, NOT PROTECTION.

+5V IS SPLIT AT FB201: +5V (buck) feeds the bead, +5V_DRV feeds U201.A1 and C201/C202/ C213 only. That containment is the point – the 72–100 mA of gate charge at 20 MHz must return inside the local 1.1 uF, not across

ai-ee
Sheet: /stage/
File: stage.kicad_sch
Title: rf-de-20m: stage – drive, LMG1020, EPC2019 pair, C_shunt trim,

Size: A2	Date: 2026-08-07	Rev: 1
KiCad E.D.A. 10.0.3		Id: 2/4



L301 AND L302 ARE ETCHED PCB AIR-CORE SPIRALS.
They are real components with real nets and NO FOOTPRINT –
P6 authors the copper. No LCSC code, excluded from the BOM.
SPIRAL-1: L +/-3%, Q >= 120 at 20 MHz, copper area >= P/7 mW.mm-2
(L301 >= 1430 mm2 at Q 100, L302 >= 950 mm2 at Q 100).
SPIRAL-5: centre-to-centre >= 38 mm, and P8 must compute the
residual mutual coupling k and fold it into the tank solve –
the two spirals are NOT independent components.
SPIRAL-6: no metal within 15 mm in plan view on either face.
A conductive plate under a spiral is a SHORTED TURN and no
copper cutout prevents it – that includes the heatsink,
brackets, standoffs and fasteners.
Zone B carries no plane on In1/In2/B.Cu by construction, and a

ai-ee		
Sheet: /tank/ File: tank.kicad_sch		
Title: rf-de-20m: tank – L_s/C_s series tank, L-match, RF output		
Size: A2	Date: 2026-08-07	Rev: 1
KiCad E.D.A. 10.0.3		Id: 2/4

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0 *(no digest recorded)*

P1 *(no digest recorded)*

P2

```
# P2 Architecture digest - rf-de-20m (20 MHz Class E GaN stage, 200 W)

**AMENDED 2026-08-07 (P2-A)** after the orchestrator read the EPC2019 datasheet
(rev (c)2021) directly and retracted two P1 figures.  This digest reflects the
amended package.

Artifacts:  ‘architecture/blocks.md‘, ‘power_tree.md‘, ‘stackup.md‘,
‘sheets.md‘,
‘constraints.json‘, ‘decisions.md‘.

---

## AMENDMENT P2-A - a retraction, plus two further errors found while
re-deriving

**RETRACTED.** ‘research/power.json‘ supplied **Coss(er) 156 pF** and **Rds(on)
22 typ / 42 max mohm**.  The datasheet publishes **no Coss(er), no Coss(tr) and
no
Eoss anywhere** - the 156 pF was invented - and Rds(on) is **36 typ / 50 max
mohm**.
The original ruling *"2 x Coss(er) = 312 pF lands on the required 317 pF, so
the
external shunt cap disappears"* is void.

**ENDORSED.** The orchestrator’s method is right:  **‘Coss(tr) = Qoss/V‘
(charge-
equivalent) is the correct basis for Class E**, because the drain waveform is
produced by integrating current into charge.  ‘Coss(er)‘ is the *hard-switching
energy* equivalent (~119 pF here, 25% low) and the 110 pF small-signal figure
is
quoted at 100 V where Coss has already fallen.  Both would have under-sized the
shunt.

**ERROR 1 - wrong voltage range.** ‘18 nC/100 V = 180 pF‘ is the average over
```

0-100 V; this drain swings to **142.5 V** and Coss keeps falling. Two independent extrapolations (power-law fit anchored on *both* published numbers, and a flat-above-100 V bound) **agree to 1.5%: Coss(tr) = 158 pF typ / 205 pF max**.

ERROR 2, the bigger one - wrong Sokal coefficient. ‘0.1836/(omega.R)‘ is the **Q_L -> infinity** constant; this board runs **Q_L = 5**, and ‘refdesign-classE-stage.json‘ D9 already said so. Sokal's published fits reproduce the fragment's power and shunt coefficients **to 0.008% and 0.13%**, confirming both. **The frozen operating point itself was computed with Q_L -> infinity constants on a Q_L = 5 design.**

	Frozen	Orchestrator's fix	**P2-A**
Vdd	40 V	37.5 V	**40 V - frozen input preserved**
R_opt	4.614 ohm	4.06 ohm	**4.13 ohm**
C_shunt required	317 pF	360 pF	**403 pF** (+38-46 pF finite-choke term)
Coss(tr) per FET	(invented 156)	180 pF	**158 typ / 205 max**
Pair supplies	312 pF	360 pF	**316 typ / 410 max**
External C0G trim	5 pF	0 pF	**87 pF, 0-133 pF of range**

The two errors pull opposite ways and the second dominates: the bus stays at 40 V, and the external trim capacitor comes back. That last point is the most valuable outcome - the original ruling treated its disappearance as elegance, but it is a **defect**: Coss spread on this part is 110-150 pF (+36%), and the external cap is the **only** absorber. At 100% device shunt a max-Coss part is unfixable without reworking etched copper.

The four rulings, as amended

1. **TWO EPC2019 - REINFORCED, and now absolute.** With the real Rds(on), a single FET reaches **Tj 160 C with a HYPOTHETICAL 0 C/W heatsink - above the 150 C absolute maximum.** There is no heatsink, TIM, via array or copper weight that saves a one-FET build (it was "138 C against a 125 C target" before). The pair:
114 C nominal / 133 C at the max-datasheet corner at theta_HS 0.7 C/W.
The "k = 2 is exactly the ZVS ceiling" sub-argument is RETRACTED (the real ceiling is 2.55) and replaced by a quantitative one:
‘P_FET(N) = 5.48/N + 2.42N + ...‘ gives **11.17 / 11.25 / 13.16 W at N = 1/2/3**.
Paralleling does not reduce dissipation at all - it halves thermal resistance.
N=3 pays +1.9 W and a three-way gate match that cannot be mirrored.
2. **C_shunt = 403 pF: 316 pF from the pair + ~87 pF external.** C203-C206 are four 1206 C0G sites, **3 populated (99 pF), no longer DNP**, in the power loop.
3. **PCB spirals, re-derived: L_s 164 nH, L_m 110 nH.** The **losses barely moved** (1000/Q and 666/Q) because ‘P = 200.Q/Q_ind‘ is **independent of R** - the tank current rises exactly as fast as the reactance falls. The lower L is geometrically helpful: 164 nH at the same OD wants a *wider* trace, which is what

Q and the thermal area want anyway.
4. **Two-zone floorplan - unchanged.** The power loop *ends* where the tank *begins*, so the two plane rules apply to disjoint regions.

New findings at P2-A

- **The frozen tank was wrong for its own Q_L.** R 4.614 -> **4.13 ohm**; C_shunt 317 -> **403 pF**; L_s 184 -> **164 nH**; C_s 447 -> **518 pF +/-5%**; L_m 115 -> **110 nH**; C_m 500 -> **530 pF**; I_tank 6.58 -> **6.96 A rms**. Tank conductor widths grow ~12% (7.2 mm, was 6.4).
- **A third P1 coefficient is refuted and NOT adopted.** The fragment's ‘C_series.omega.R = 0.63467‘ implies a net series reactance of **3.4 R** - not a
Class E network. Sokal's fit gives **0.26906 -> 518 pF and 1.283 R**, which also
agrees within 3.5% with the independent ‘X_net = 1.1525 R‘ relation.
SIM-4 gained a measure (‘x_net_series_over_r‘) specifically to arbitrate it.
- **Vdd is NOT a ZVS knob - but it IS a free thermal derating knob.** ZVS in Class E
is a property of the *network*: the equations are linear in Vdd and the ZVS/ZdVS
conditions constrain only the current waveform shape. So the bus can be backed down at bring-up without losing ZVS: **40 V/200 W/Tj 133 C -> 36 V/162 W/Tj 119 C**
at the max corner, zero rework. This is the mitigation for the compounded worst
case (170 C), and it corrects the orchestrator's framing - lowering the bus alone
cannot fix a shunt-capacitance error.
- **The gate-loop budget TIGHTENED to 0.48 nH per FET** (was 0.84). The datasheet's
Ciss 200 pF / Crss 0.7 pF give C_GS ~ 199 pF, not the ~350 pF the retracted Qg implied. **Now the tightest layout spec on the board**, with a stated fallback (R_G 3 ohm -> 0.84 nH, at +0.3 W turn-off loss).
- **The real Qg of 1.8 nC IMPROVED the +5 V rail**: 143 -> **99 mA**, 3.0x headroom,
and gate-resistor dissipation 0.047 -> 0.029 W per part.
- **HS-1 tightened: theta_HS <= 0.7 C/W** (was 1.4). A passive bolt-on will not do it.
- **BOM +$4/board**: EPC2019 is **OUT OF STOCK at LCSC (stock 0), repriced $2.17 -> $3.93**. Owner approved continuing and **holding the order**; P10 re-verifies. Delivered estimate **~$32-40/board (~$160-200 for 5)**.

Unchanged

Stackup ‘JLC04161H-1080B‘ (chosen for its 0.2444 mm L1-L2 gap), 1 oz outer, TG155 + ENIG + POFV, 100 x 80 mm with pre-authorised growth to 120 x 80, the two-zone floorplan and plane regions, the relaxed controlled-Z output (D7), DC-coupled unipolar drive (D8), no protection parts, three sheets and the net-naming contract. **Vds,pk stays 142.5 V** - it depends only on Vdd.

Pipeline traps - two carried, one retired

1. **‘rules_gen‘ only solves impedance for DIFFERENTIAL pairs.** A single-ended
‘high_speed‘ entry with ‘impedance_ohm: 50‘ emits **no width rule at all**.
‘/tank/RFOUT‘ is declared under ‘power‘ instead - which is also the right physics
(**OPEN-2**).
2. **‘detect_diff_pairs‘ auto-pairs ‘high_speed‘ nets ending ‘_H‘/‘_L‘**, hence

‘/stage/GATE_ON‘ / ‘_OFF‘.
3. **‘planes_gen‘ has no void or keepout support**; zone B is unpoured by
construction and the spiral courtyards need hand-added rule areas at P6.
‘board_init‘ does not origin the outline at (0,0) - translate every rect by
‘outline_bbox‘ at P5.
4. **RETIRED:** "‘rules_gen‘ puts every power net in ONE class at the widest
width"
is **fixed** in the current script (one class per required width).

Riskiest decision

**Coss hysteresis is now the dominant FET loss term (4.83 W of 11.25 W) and
rests
on an assumed 10% of stored energy that EPC does not publish** - the same class
of
gap that produced the retracted Coss(er). At 15% the pair reaches ~13.7 W and
Tj
~127 C. Mitigations are in the architecture at zero cost: the trim bank, the
bus-derating ladder, and a first bring-up step that settles it in an afternoon
(measure DC input power at 200 W out, thermal-image both dies).

‘constraints.json‘

Lint-clean: 0 errors, 0 warnings. 4 high_speed, 6 power, explicit empty
diff_pairs, 5 voltages, 3 voltage_pairs, 7 thermal, 6 planes, placement
edges/groups/keepouts/separation/fixed. Every current, voltage and power
figure
re-derived at R = 4.13 ohm.

Sim gate - SEVEN benches (SIM-7 added at P2-A)

SIM-1 ‘classe_zvs_nominal‘ (Vds@turn-on <= 5 V, dVds/dt +/-0.2 V/ns, Pout
190-210 W, Vds pk <= 155 V). **SIM-2** ‘cshunt_sweep‘ - **now validates the
whole
amended basis**, with ‘coss_tr_per_fet_pf‘ bounded to **140-180 pF** from the
vendor
nonlinear model plus a 300-430 pF sweep and ‘trim_pf_needed <= 133‘; must pass
before P4 fixes any tank value. **SIM-3** ‘gate_symmetry‘ (Id imbalance <= 15%
at
+/-0.15 nH, Vgs inside -4.0/+5.75 V). **SIM-4** ‘tank_match‘ (Zin 3.72-4.54
ohm,
H2 <= -20 dBc, H3 <= -33 dBc, **plus ‘x_net_series_over_r‘ in 1.0-1.6 to
arbitrate
C_s**, and the spiral-to-spiral mutual k). **SIM-5** ‘loss_split‘ (pair <=
13.0 W
typ / <= 16.0 W max-corner - the numbers the 0.7 C/W heatsink was solved
against).
SIM-6 ‘gate_loop‘ (advisory). **SIM-7** ‘bus_derating‘ **(new)** - re-run
SIM-1
at 40/38/36/34 V with the network unchanged, to confirm ZVS is Vdd-independent
and
the derating ladder is real.

P3 *(no digest recorded)*

P4 *(no digest recorded)*

Gate attempt history

(no entries)

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-07T22:07:56	Architecture approved incl. override of frozen single-FET spec to 2x EPC2019 parallel; unipolar 0/+5V drive; controlled-Z relaxed; outline 100x80 with pre-auth growth to 120x80
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
`state.json`	13,328 B	
`kicad/rf-de-20m.kicad_sch`	7,267 B	
`brief/brief.md`	4,153 B	
`requirements.md`	16,663 B	
`architecture/blocks.md`	37,342 B	
`architecture/constraints.json`	24,610 B	
`architecture/decisions.md`	30,793 B	
`architecture/power_tree.md`	17,822 B	
`architecture/sheets.md`	14,529 B	
`architecture/stackup.md`	18,766 B	
`reports/review-schematic.json`	7,817 B	
`reports/review-schematic.md`	20,810 B	
`reports/schematic.pdf`	232,266 B	
`reports/top.net`	129,049 B	